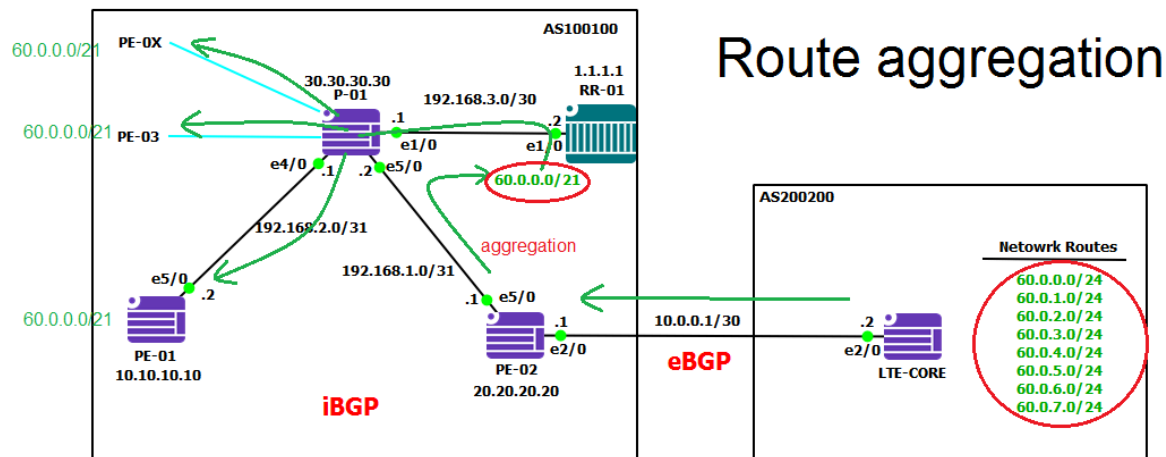


Advanced IPv4 Addressing & Network Design — Consolidated Technical Briefing

All 16 of the Possible /20 Networks for 10.0.*.*

Network Address	Usable Host Range	Broadcast Address:
10.0.0.0	10.0.0.1 - 10.0.15.254	10.0.15.255
10.0.16.0	10.0.16.1 - 10.0.31.254	10.0.31.255
10.0.32.0	10.0.32.1 - 10.0.47.254	10.0.47.255
10.0.48.0	10.0.48.1 - 10.0.63.254	10.0.63.255
10.0.64.0	10.0.64.1 - 10.0.79.254	10.0.79.255
10.0.80.0	10.0.80.1 - 10.0.95.254	10.0.95.255
10.0.96.0	10.0.96.1 - 10.0.111.254	10.0.111.255
10.0.112.0	10.0.112.1 - 10.0.127.254	10.0.127.255
10.0.128.0	10.0.128.1 - 10.0.143.254	10.0.143.255
10.0.144.0	10.0.144.1 - 10.0.159.254	10.0.159.255
10.0.160.0	10.0.160.1 - 10.0.175.254	10.0.175.255
10.0.176.0	10.0.176.1 - 10.0.191.254	10.0.191.255
10.0.192.0	10.0.192.1 - 10.0.207.254	10.0.207.255
10.0.208.0	10.0.208.1 - 10.0.223.254	10.0.223.255
10.0.224.0	10.0.224.1 - 10.0.239.254	10.0.239.255
10.0.240.0	10.0.240.1 - 10.0.255.254	10.0.255.255



1. Big Picture: Why This Topic Matters (GATE Perspective)

This briefing sits at the **peak of IPv4 addressing difficulty**—where **CIDR, subnetting, VLSM, and ISP-level hierarchical allocation intersect**.

Historically, GATE uses this zone to test:

- Mathematical rigor (powers of two, prefix derivation)
- Logical filtering (valid vs invalid CIDR blocks)
- Design thinking (how ISPs minimize routing entries)

Your document correctly identifies **hierarchical ISP allocation** as a *high-probability 2-mark question*, especially because:

- It is standard in textbooks (e.g., Forouzan)
- It has been under-tested recently
- It combines **CIDR + aggregation + arithmetic sanity checks**

2. The “AD Rule 2.0” — Why It’s So Powerful

Core Idea

Instead of *starting from the network*, start from the **host requirement**.

[
$$\text{Prefix} = 32 - \lceil \log_2(\text{Addresses Required}) \rceil$$

]

This turns multi-step subnetting into a **5-second mental operation**.

Why It Works

- CIDR blocks must be powers of two
- Host bits = smallest power of two that satisfies demand
- Prefix is simply the complement

Practical Mapping (High-Yield)

Addresses Needed	Host Bits	Prefix
4	2	/30
16	4	/28
64	6	/26
128	7	/25
256	8	/24
1500	11	/21

Exam shortcut:

Never calculate Network ID first. Always calculate Host Bits first.

3. Validating a CIDR Block: The Non-Negotiable Test

Once the prefix is known, **only two checks matter**.

Check 1: Range Check

Is the candidate block **inside** the ISP's allocated supernet?

Example:

- ISP range: `202.61.0.0/17` → third octet `0–127`
- Any block starting at `202.61.144.x` immediately rejected

Check 2: First-IP Divisibility Rule (Most Important)

If block size = (2^H) , then:

The first IP must be divisible by (2^H)

Binary meaning:

- Last **H bits must be all zero**
- That IP automatically becomes a valid **Block ID**

GATE 2020 Insight

For a `/21` block:

- Host bits = 11
- Block size = 2048
- Valid third-octet starts: **0, 64, 104, ...**

This single rule eliminates **75% of options instantly**.

4. Hierarchical ISP Allocation — The Real Design Problem

This is where **CIDR stops being arithmetic and becomes architecture**.

Given

ISP owns:

`162.72.0.0/16` → 65,536 IPs

Customer Groups (Structured Allocation)

Group	Customer s	IPs per Customer	Customer Prefix	Group Size
G1	128	256	/24	2^{15}
G2	128	64	/26	2^{13}

G3

64

128

/25

2^{13}

Key Observations

- Each group is internally **uniform** → enables aggregation
- Groups themselves are powers of two → allows clean slicing
- ISP can advertise **one prefix per group**, not per customer

Remaining Address Calculation (Clean Power-of-Two Math)

[
 $2^{16} - (2^{15} + 2^{13} + 2^{13})$
]

[
 $= 2^{16} - (2^{15} + 2^{14})$
]

[
 $= 2^{14} = 16,384 \text{ IPs}$
]

Design takeaway:

Proper hierarchy doesn't just save IPs—it saves **routing table entries**.

5. Why This Is a Classic GATE Trap

GATE often mixes:

- **Correct prefix**
- **Incorrect starting IP**

Students stop after finding /21, but forget:

- CIDR blocks must be **aligned**
- Alignment is enforced by **divisibility**, not intuition

That's why options like:

- 202.61.84.0/21 ❌

- 202.61.104.0/21 ✓

appear together.

6. Conceptual Closure: What You've Actually Finished

With this chapter, you've covered:

- ✓ Classful addressing limitations
- ✓ CIDR fundamentals
- ✓ Subnetting & VLSM
- ✓ Supernetting & aggregation
- ✓ ISP-level hierarchical allocation
- ✓ Exam-grade shortcut logic

This is the **end of IPv4 addressing as a concept**.

Everything beyond this (Error Control, Flow Control) shifts from **addressing logic** to **protocol behavior**.